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Hybrid Recommender System for Grocery Retail: Leveraging Frequent Pattern Mining and Collaborative Filtering with Recency-Aware Personalization

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Table of Contents

Contents

1	Introduction	3
2	Related Work	3
3	Exploratory Analysis	3
3.1	Dataset description	3
3.2	User Behavior Analysis	4
4	Implementation	5
4.1	Task 1: Frequent pattern mining	5
4.1.1	Methodology for selecting appropriate metric and algorithm	5
4.1.2	FP-Growth Algorithm	6
4.2	Task 2: Collaborative filtering	7
4.2.1	Collaborative Filtering Training	8
4.2.2	Collaborative Filtering model evaluation	8
4.3	Task 3: System integration	9
4.3.1	Literature review	9
4.3.2	Practical implications for our work	9
4.3.3	Integration code	9
4.4	Experimental evaluation.....	11
4.4.1	Program implementation.....	12
4.4.2	Evaluation Results and Discussion.....	12
5	Conclusion and future work	12
	Appendix	14
A	Collaborative Filtering Experiments	14
	Contributions and Reflections	15
A.1	Task 1.....	15
A.2	Task 2.....	16
A.3	Task 3.....	17

Executive Summary

The objective of this report is to outline the development and evaluation of a recommender system, which recommends personalized grocery items based on customers' purchase histories to boost sales. Our developed system comprises two main components: **frequent pattern mining** and **collaborative filtering**, in combination with **recency-aware personalization** to improve recommendation quality. In this work, we explore how **mining frequent patterns** can generate tailored recommendations based on user preferences and behaviour. The report concludes with suggestions for further tuning, including potential improvements in scalability and personalization that meet real-world criteria.

1 Introduction

Recommender systems have long been recognized as a widespread method across the world, with prominent applications in major commercial platforms such as Amazon, Netflix, and Spotify [1]. These systems play a key role in enhancing customer experience through personalized recommendations that raise the level of user engagement and sales. According to Falk, a recommender system calculates and provides relevant content to the user based on knowledge of the user, content, and interactions between the user and the item [2].

2 Related Work

Over the past decades, researchers have studied and developed recommender systems in several ways. Specifically, combining frequent pattern mining with collaborative filtering has been extensively investigated in recommender systems to use both transactional patterns and user-item interaction data for enhanced recommendation accuracy [3]. Other research has introduced hybrid models that jointly utilize pattern mining and collaborative filtering to address challenges such as data sparsity and cold-start problems, achieving better performance than either method alone [4].

In this paper, we not only utilize the common methods of pattern mining and collaborative filtering but also seek to fill the research gap in this area by introducing the Recency-Aware Personalization approach.

3 Exploratory Analysis

3.1 Dataset description

Our research utilized a grocery retail transactional dataset that has been specially structured for the purpose of developing and assessing recommendation systems. The entire dataset comprises approximately 38,600 records, which is arranged chronologically by transaction date. The dataset is split into two identical segments: the training set, which comprises the initial portion of approximately 19,300 records, and the test set, which includes the latter portion containing roughly 19,300 records.

Each transaction record includes several attributes that are largely self-explanatory, such as:

- **User_id:** A unique identifier for each customer.
- **itemDescription:** The description of the purchased item.
- **Date:** The purchase date of the transaction.
- **year, month, day:** Decomposed date components consistent with the Date column.
- **day_of_week:** An integer encoding where Monday = 1 and Sunday = 7.

3.2 User Behavior Analysis

The exploratory data analysis reveals important patterns in user behavior and item popularity. Fig. 1 shows that the daily transaction numbers are relatively stable, with most of the days having a total of 15 to 35 purchases.

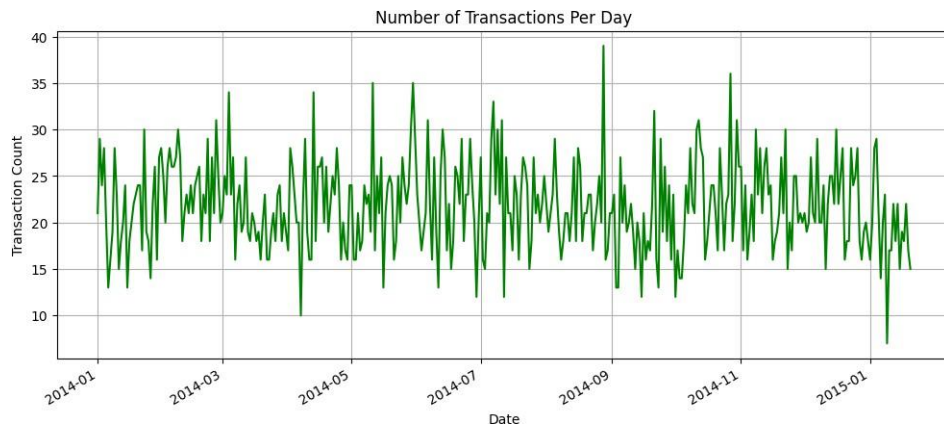
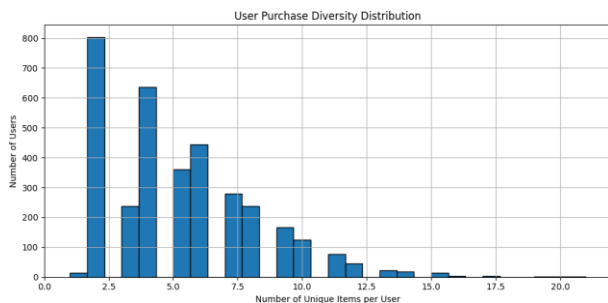
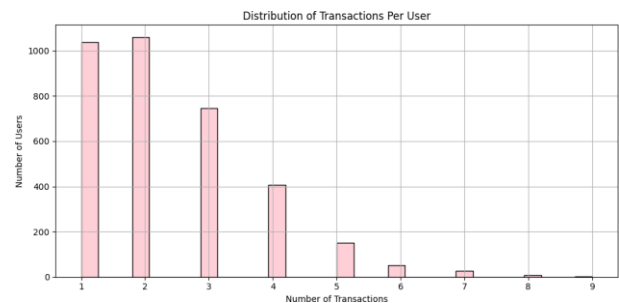


Figure 1: Daily purchase

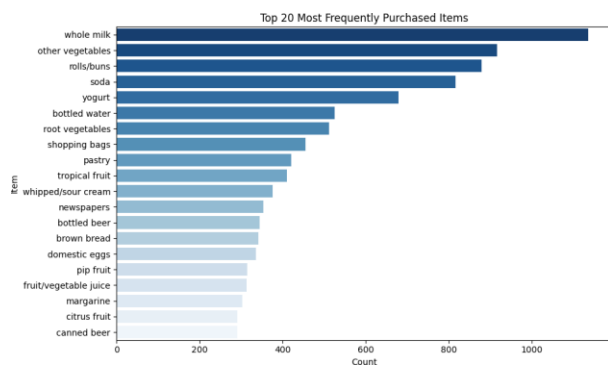
However, Fig. 2b indicates that the majority of customers have made only one or two transactions, and the purchasing diversity chart in Fig. 2a shows that over 800 users bought only two unique items, which indicates that the majority of users are only interacting with a limited number of items. This sparsity might pose a significant challenge for collaborative filtering, as collaborative filtering relies on shared behavior across users [5].



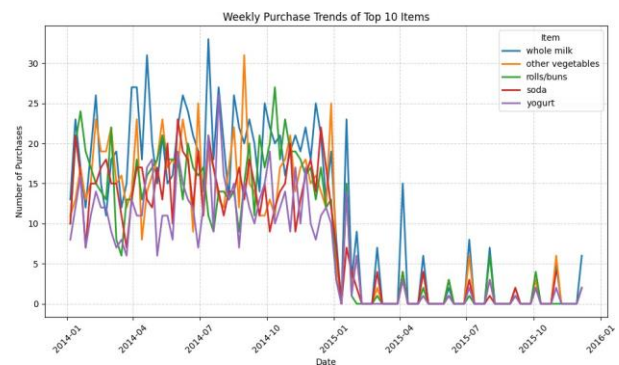
(a) Unique items per user



(b) Transactions per user



(c) Top 20 purchased items



(d) Weekly trend of top items

Figure 2: User behavior and item popularity analysis

When looking at popular items, items like whole milk, other vegetables, rolls/buns, soda, and yogurt are the most popular items. Fig. 2c and these products have consistent weekly demand, which is followed by a sudden decline post 2015. Insights from the analysis indicate that the users are repeat purchasing certain items, there is minimal overlap between user purchases, while certain products exhibit consistent demand.

4 Implementation

4.1 Task 1: Frequent pattern mining

4.1.1 Methodology for selecting appropriate metric and algorithm

In this section, we justify our choice of the FP-Growth algorithm as the optimal model to mine frequent itemsets along with support metric.

Input: Training set containing transactional data.

Output: A list of top 5 frequent itemsets, which is ranked by recency-weighted scores.

Threshold Selection Reasoning: To establish the best support threshold for FP-Growth, we evaluated the number of patterns generated at different thresholds (Fig. 3). A threshold of 0.001 generated over 500 patterns, while also adding noise to the analysis by generating many infrequent itemsets. Other thresholds below 0.005 resulted in too few patterns (less than 50) and therefore missing meaningful co-purchase behaviors that exist in grocery retail. Our selected threshold of 0.005, resulting in 100 patterns approximately, justifies the threshold by balancing the number of missed patterns whilst reducing noise. It is also important to ensure computational efficiency for the case study, as well as ensuring trustworthy recommendations to grocery customers when benchmarking parameters, as Han et al. (2011) suggest not to go too low (as it adds noise) or too high (as it will lose some key patterns) [6] and Aggarwal (2016) recommends balancing threshold risk in terms of retail scenario [7].

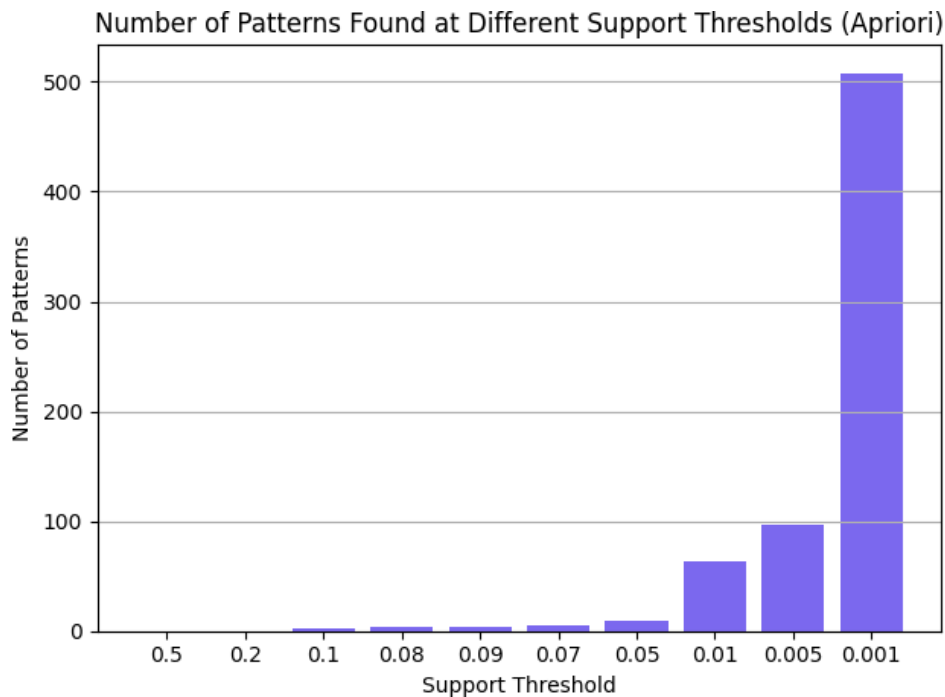


Figure 3: Threshold experiments

Experimentations and Evaluation: We conducted several experiments comparing Apriori and FP-Growth with different configurations. This includes setting with and without recency weighting, with the chosen threshold 0.005 that has already been justified above. The metrics we used in this work are runtime and memory usage, which represent computational efficiency, as depicted in Table 1.

- **Observations:** We chose FP-Growth with Recency as our final model because it gave us a good balance between performance and efficiency. According to the table, it had the lowest memory usage (16.88 MB) and almost the same runtime (1.88s) as regular FP-Growth. What made it unique was that we were able to use recency scores to increase the relevance of our recommendations by reflecting on the age of each item purchased whilst also maintaining speed and memory. Not only was it more memory efficient than Apriori models, but it was also significantly quicker, while producing results with comparable high precision. Thus, it was the most efficient practical and useful model for our analysis.
- **Scalability and Efficiency Advantage:** While FP-Growth's runtime is high compared to previous models, it is expected to perform better in terms of memory efficiency. Memory efficiency is important for our grocery retail recommender, because of the nature of the potentially large transactional datasets and ability to scale down the memory overhead; especially as grocery retail has dense datasets with frequent repeated purchases, such as whole milk as shown in Fig. 2c) [8]. To address the high runtime issue, we provided frequent patterns and the collaborative filtering user-item matrix in advance using an eager approach to all the user interactions, thus minimizing the impact of runtime and providing real-time recommendation without any change in user interaction.
- **Recency-Aware Personalization Benefit:** Finally, FP-Growth + Recency aligned with our data mining purpose of recency-aware personalization, as recency can prioritise recent user behavioural data with shared value (increasing similarity) to promote relevancy in a recommendation context, an idea supported by Koren et al. (2009) who concluded that the recent interactions are a better representation of the user's current preferences [9]. It is apparent, in terms of memory efficiency, scalability of recency, FP-Growth + Recency was ultimately determined to be the best model for frequent pattern mining in our hybrid recommender system.

Table 1: Computational Performance of FP-growth vs. Apriori with and without Recency

Algorithm	Support Threshold	Runtime (s)	Memory Usage (MB)
Apriori	0.005	0.08	66.20
FP-Growth	0.005	1.90	16.89
Apriori + Recency	0.005	0.07	66.20
FP-Growth + Recency	0.005	1.88	16.88

4.1.2 FP-Growth Algorithm

The FP-Growth algorithm provides an alternative way to mine a frequent itemset by compressing transaction records using a special graph data structure called FP-Tree. FP-Tree can be thought of as a transformation of the dataset into graph format. Rather than the generate and test approach used in the Apriori algorithm, FP-Growth first generates the FP-Tree and uses this compressed tree to generate the frequent itemsets. The efficiency of the FP-Growth algorithm depends on how much compression can be achieved in generating the FP-Tree [6]. Mathematically, the conditional pattern

base (**CPB**, i.e., *Conditional Pattern Base*) for an item i is derived from the FP-tree to recursively generate conditional FP-trees:

$$\text{CPB}(i) = \{\text{prefix path of } i \mid \text{frequency}(i)\} \quad (1)$$

FP-Growth significantly reduces computational overhead and is highly effective for dense datasets and large-scale transactional data, making it suitable for practical recommender system applications [8]. FP-Growth employs a **depth-first search** algorithm to explore frequent itemsets recursively and efficiently [6].

Algorithm 1 FP-Growth Algorithm

Require: Dataset D , minimum support threshold σ

Ensure: All frequent itemsets

- 1: Construct the FP-tree from D
 - 2: **for all** items i in FP-tree (in support ascending order) **do**
 - 3: Construct conditional pattern base for i
 - 4: Construct conditional FP-tree from conditional pattern base
 - 5: Recursively mine the conditional FP-tree
 - 6: **end for**
 - 7: **return** All frequent itemsets
-

4.2 Task 2: Collaborative filtering

Collaborative filtering (CF) is among the most widely used methods in recommender systems since users with similar past behaviours or preferences are likely to show similar behaviours in the future [10]. Our hybrid recommender system's collaborative filtering component uses cosine similarity metrics and the K-Nearest Neighbours (KNN) item-based method. This approach finds relationships between grocery products by means of collective user purchase patterns, allowing customized recommendations outside of simple associations.

Algorithm 2 Item-based Collaborative Filtering using KNN

Require: Training dataset D , user ID u , number of neighbors k

Ensure: Top-N recommended items

- 1: Build user-item matrix from D
 - 2: Train KNN model on user-item matrix with cosine similarity
 - 3: **if** u not in user-item matrix **then**
 - 4: **return** Empty recommendation list
 - 5: **end if**
 - 6: Retrieve items previously interacted by user u
 - 7: **for all** item i in user's interacted items **do**
 - 8: Find k nearest neighbors for item i
 - 9: **for all** neighbor item j not interacted by user **do**
 - 10: Accumulate similarity score for item j
 - 11: **end for**
 - 12: **end for**
 - 13: Rank candidate items by cumulative similarity scores
 - 14: **return** Top-N highest-ranked items
-

4.2.1 Collaborative Filtering Training

Sparsity problem User-based collaborative filtering requires a user-item matrix. However, the user-item matrix was highly sparse, with 98.86% of the matrix having 0 elements, i.e., 3.33% of users have overlapping purchase history. Such sparsity complicates identifying user neighbors [2]. To address this item, based collaborative filtering was used, which is robust in sparse settings [11].

The train set was split into training and development sets with an 80-20 split. The collaborative filtering was developed by constructing a user-item matrix where each row represents an item and the column represents the user on the training set. The matrix entries have non-zero values that indicate the number of times an item was purchased by a user. A zero value means that the item was not purchased by the user. A K-nearest neighbors model was employed using cosine similarity as the distance metric and a brute-force search algorithm. This configuration was selected post experimenting with different metrics such as Jaccard similarity with binary user-item matrix and user-item matrix with frequency of purchase and recency factored (appendix A). The best configuration was then evaluated using the dev set and then chosen based on precision@k and recall@k ($k = 5$) scores. Here precision and recall are chosen to understand how accurate and complete the recommendations are, helping evaluate both the quality and coverage of the system [12].

The best-performing model was the frequency-aware item-user matrix with cosine similarity, achieving a precision of 0.0938 and a recall of 0.2082.

4.2.2 Collaborative Filtering model evaluation

Two main recommendation strategies were evaluated. A standard user-item-based collaborative filtering model and a hybrid model that utilizes frequent patterns from frequent itemset mining. The CF model relies only on item-based similarity, which was derived from historical interactions, to make recommendations. In contrast, the hybrid strategy combines the CF-based recommendation with results from mining frequent patterns, which identifies the common co-purchased items that appear in the dataset. The model filters out items already seen by the user and prioritizes those that appear in both the user's neighborhood and the globally frequent itemsets and de-prioritizes items with lower confidence.

Model	Precision@5	Recall@5	Hit Rate@5	NDCG@5
Collaborative Filtering (CF)	0.0848	0.1558	0.3545	0.1387
Hybrid Model	0.112	0.2045	0.4627	0.1679

Table 2: Performance Comparison of CF and Hybrid Models (Top-5 Recommendations) on development set

The models were trained on the training set and evaluated on the development set. The model performance was compared using Precision@k, Recall@k, Hit Rate@k, and NDCG@k ($k = 5$).

Precision@k and Recall@k were chosen to evaluate the overall relevance of the recommendation and if the model retrieved all possible items, respectively. Hit Rate and NDCG were selected to check if the user received one relevant item and NDCG@k for the ranking quality, which rewards relevant items that appear early on the recommendation list [13].

The results showed that the hybrid model outperformed the standalone CF model, achieving higher scores across all metrics as presented in Table 2, which indicates improved recommendation relevance by leveraging personalized recommendation and global item patterns.

The hybrid model performed better than the standard model due to the diverse purchase patterns among customers shown in Fig. 2a. Since most users tend to interact with a small number of unique

items, the collaborative filtering model has difficulty determining the neighbors. The hybrid model mitigates this by using globally frequent itemsets, which helps in enhancing the recommendations even when user histories are sparse or repetitive.

4.3 Task 3: System integration

4.3.1 Literature review

Recommendation systems have historically leveraged frequent pattern mining to represent co-occurrence relationships from transactional data. When discussing frequent itemsets, Aggarwal (2016) argues frequent itemsets can overcome sparsity by identifying global patterns to augment user-based recommendations [7]. Consider a market basket analysis for example, if the frequent itemset bread, butter was identified, the frequent pattern could suggest the user buy "butter" with bread, which might be valuable to a user with limited purchase history; in this case, the corresponding recommendation would not be user-based, but still able to suggest relevant or complimentary items.

Lin et al. (2002) introduced a hybrid model that integrates FP-growth and collaborative filtering. They found that the model produced a relatively accurate recommendation system in retail contexts. They also found that frequent patterns were especially useful in real-world situations, especially if a user was in a cold-start problem [14].

Koren et al. (2009) advanced and expanded the discussion of time to recommendation systems, arguing that users' preferences change over time and that recent interactions are normally more indicative of the user's current preference. Their approaches utilizing time-awareness demonstrated that they could improve recommendation accuracy by modeling recency and temporal trends directly in collaborative filtering algorithms [9].

4.3.2 Practical implications for our work

- **Hybrid approach:** Inspired by the work of Lin et al., we decided to use frequent itemsets to boost collaborative filtering recommendations. This helps us to combat the sparsity issue in our dataset and provides fallback for new users.
- **Recency weighting:** We apply the concept of recency-aware personalization and temporal dynamics that was highlighted by the work of Koren et al. (2009), to weight recent user feedback higher than feedback that is less recent.

4.3.3 Integration code

The integration code merges collaborative filtering and frequent pattern mining to provide personalized recommendations. It processes the transactional dataset by first preparing the user-item interactions and mining the frequent itemsets, accounting for their recency so that we can provide recommendations based on recent user activity. A user interface has been designated for entering the User ID and selecting the recommendation method (using frequent patterns or only collaborative filtering). If the user does not have a history of activity in the dataset, we derive recommendations directly from the itemsets since collaborative filtering alone cannot handle new users. The implementation also ensures the computations are efficient and that the models are pre-built in eager approach. The system is then evaluated on test set to understand overall performance using similar metrics that were used to evaluate development set.

Fig. 4 outlines the architecture of the hybrid recommender system which loads the model in eager approach, confirming how historical transactions as data are processed and in turn, frequent itemsets

are created through FP-growth and a user-item matrix to create collaborative filtering, and then implemented into the recommendation system to process user input and ultimately provide recommendations. Fig. 5 contains the flow of the hybrid recommender system depicting the decision-making process the system takes, wherein it decides whether to use frequent itemsets, next creating frequent patterns, generating recommendations from collaborative filtering, and lastly, re-ranking the items from overlapping frequent patterns to increase relevance.

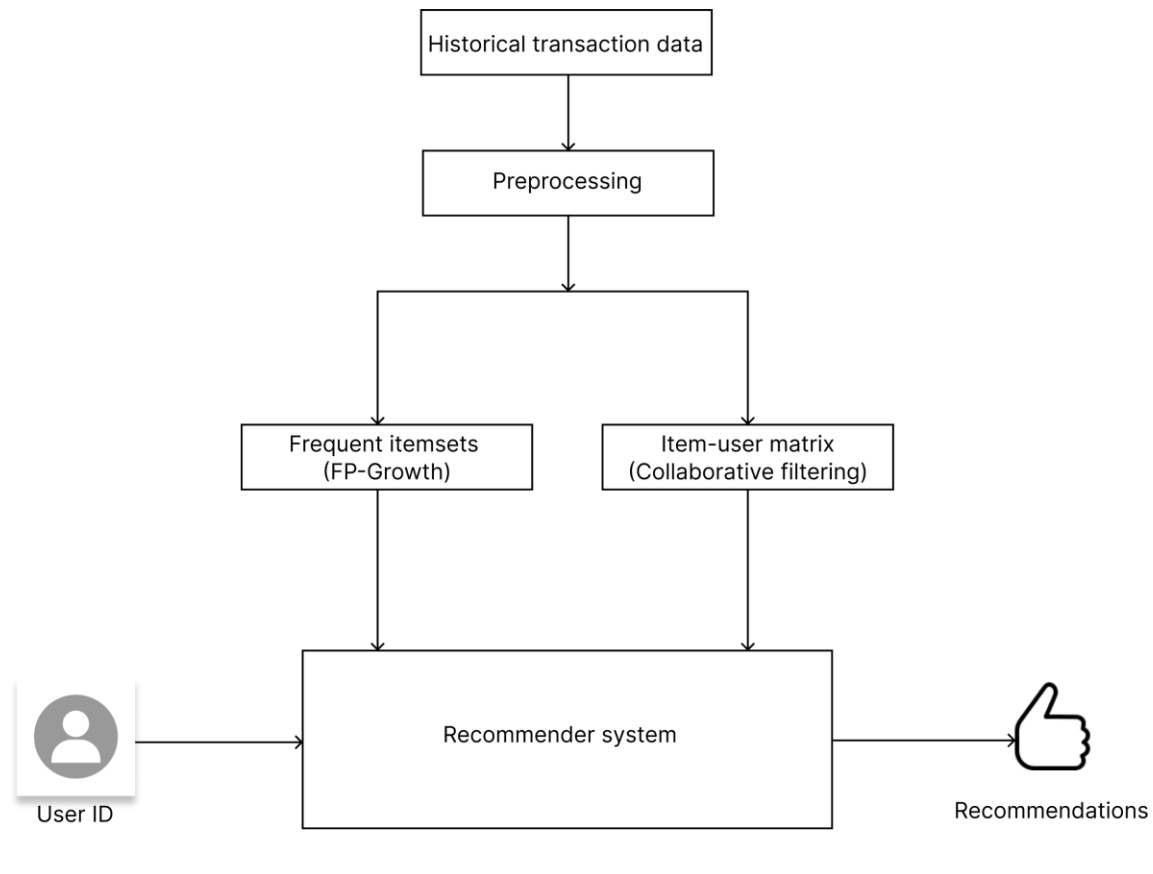


Figure 4: Architecture of the eager hybrid recommender system

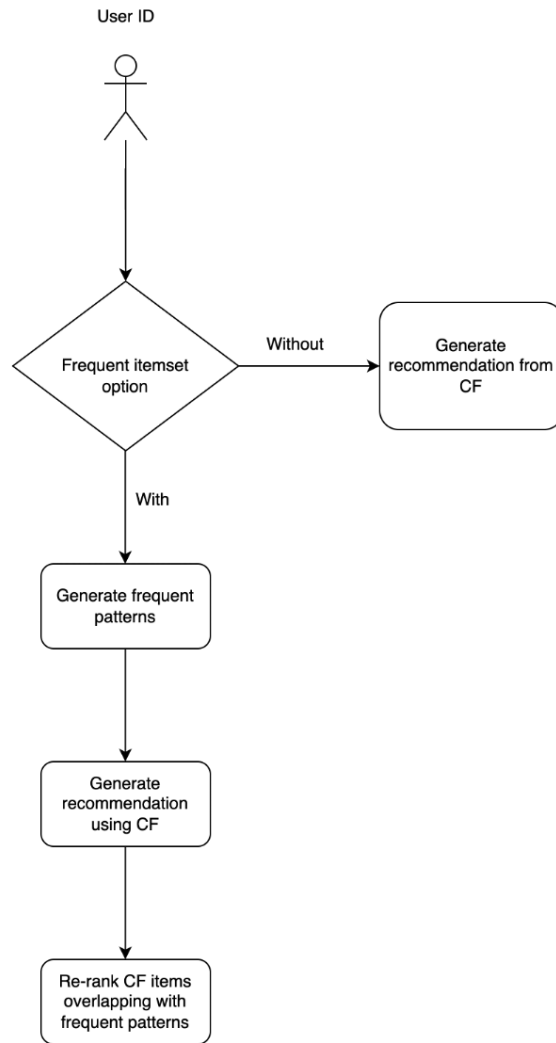


Figure 5: Flow diagram for hybrid recommender system

4.4 Experimental evaluation

User ID:

Method:

Generating recommendations for User 2351 using frequent itemsets...

Top 5 Recommendations for User 2351:

1. whole milk
2. other vegetables
3. soda
4. yogurt
5. rolls/buns

User ID:

Method:

Generating recommendations for User 2351 without frequent itemsets...

Top 5 Recommendations for User 2351:

1. soda
2. other vegetables
3. whole milk
4. rolls/buns
5. yogurt

Figure 6: Comparison of recommendation outputs with and without frequent itemsets for User ID 2351

4.4.1 Program implementation

To demonstrate the system’s functionality, we tested it using both methods with arbitrary User ID (2351) from the dataset. After running different test cases, we received outputs displaying in a simple User Interface created by Jupyter Widgets, as shown in Fig. 6. These examples show that incorporating frequent itemsets can adjust the recommendation order, prioritizing items that align with frequent patterns, while still being relevant to the user’s transaction history.

4.4.2 Evaluation Results and Discussion

For the purpose of examining the performance of our hybrid recommender system, we evaluated the hybrid model (which utilizes frequent itemsets) and the stand-alone collaborative filtering (CF) model (that does not utilize frequent itemsets) against the test set. We assessed each model’s performance with metrics of Precision@k, Recall@k, Hit Rate@k, NDCG@k ($k = 5$), runtime and peak memory, summarized in Table 3.

Table 3: Performance Comparison of Hybrid and CF Models on Test Set

Model	Precision@5	Recall@5	Hit Rate@5	NDCG@5	Runtime (s)	Peak Memory (MB)
CF (Without Frequent Itemsets)	0.2014	0.1886	0.6476	0.2480	89.66	26.48
Hybrid (With Frequent Itemsets)	0.2040	0.1911	0.6529	0.2363	102.05	26.43

It can be clearly seen from the Table 3 that the hybrid model outperforms the standalone CF model with respect to most metrics (refer to the bold figures in the table that represents better metrics). We observed slight improvements in the metrics for Precision@5, Recall@5, and Hit Rate@5, these improvements were 1.29%, 1.33%, and 0.82%, respectively. Although these enhancements were modest, they still highlight the potential of leveraging frequent itemsets in order to mitigate the sparsity issue commonly observed in CF systems [7]. However, it is noticeable that the stand-alone CF model had better performance than the hybrid model regarding the NDCG@5 metric (0.2480 vs. 0.2363). This indicates a better structure of recommendations based purely on CF similarity metrics [9]. The hybrid model was running slower than the CF model for the runtime measure (89.66s vs. 102.05s), which indicates the computational overhead introduced by frequent itemset mining. [6]. Consequently, the hybrid model required less memory (26.43 MB) to perform its recommendations, compared to the memory usage of the CF model (26.48 MB), this suggests that incorporating frequent itemsets makes the system more memory-efficient. This finding has already been emphasized by previous studies, especially the work of Zaki et al. (2014), which mentioned the memory benefits of compact pattern mining algorithms such as FP-Growth. [8].

5 Conclusion and future work

This paper describes the implementation of our hybrid recommender system that employs hybrid approach, which comprises two main components: frequent pattern mining and collaborative filtering, in combination with recency-aware personalization to improve recommendation quality, also highlighting both its strengths and limitations. However, we identified an issue that could significantly hinder the real-world recommendation performance, especially in resource-constrained environments: the computational overhead that our system faced. For future work, we plan to optimize runtime efficiency and refine recommendation ranking accuracy to make our recommender system adapt to real-world market needs.

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Appendix

A Collaborative Filtering Experiments

Model	Precision@5	Recall@5
Binary Purchase Matrix – Jaccard Similarity	0.0926	0.2053
Frequency-Aware Item user Matrix – Cosine Similarity	0.0938	0.2082
Recency and Frequency Weighted Matrix – Cosine Similarity	0.0621	0.1376
Item-Based CF with Recency-Based Weighting – Cosine Similarity	0.0337	0.0713
User based CF – Jaccard Similarity	0.0366	0.0654

Table 4: Performance Comparison of Recommendation Models (Top-5 Recommendations)

